

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Geology and Geophysics

2. **Subject Name:** Seismology **L-T-P: 3-1-0**
Credits:4

3. **Pre-requisites:** None

4. **Syllabus and reference books:**

Syllabus: Classical and Mathematical Seismology: Continuum. Elastic deformation and stress tensors and their relationship, Helmholtz's Theorem. Equation of motion in an elastic medium. Generalized Hooke's Law. Wave surfaces, rays and their properties. Anisotropy and its effects on rays. Rotational, dilatational, irrotational and equivoluminal waves. Velocities and motions associated with body waves. Amplitude and energy transport. Reflection and refraction of elastic waves at solid-solid and liquid-liquid interfaces. Head waves. Inhomogeneous & Evanescent wavefronts and bounded waves. Development of Wave equation solutions in terms of rays & Eikonal equations. Equation of travel time when velocity increases continuously with depth. Surface Waves: Rayleigh and Love waves.

Earthquake Seismology: **Introduction:** Historical development of seismology. Seismic sources, earthquake sources involving shear faulting. Quantification of earthquakes. Earthquake distributions. Global faulting patterns and rupture models. Internal structure of the earth. The framework and mechanism of plate tectonics, ocean ridges and continental rifts, transform and transcurrent faults, subduction zones, mountain ranges. **Seismometry:** Inertial pendulum systems. Electromagnetic instruments. Global Networks. Force-feedback instruments and digital global networks. Seismic arrays and regional networks. **Seismogram interpretation:** Nomenclature. Traveltime curves, locating earthquakes by single and multi station and inverse methods. **Determination of earth structure:** Seismic earthquake tomography. Joint hypocenter determination (JHD). **Seismic sources:** Faulting sources, elastostatics, double couple hypothesis, elastodynamics. Seismic moment tensor. Focal mechanisms and fault plane solutions. **Earthquake kinematics and dynamics:** One dimensional Haskell source model, source spectrum, stress drop and rupture velocity. Magnitude scales, seismic energy and magnitude. Aftershocks and fault area. Earthquake statistics. **Seismotectonics:** Divergent, transcurrent and convergent boundaries. Intraplate earthquakes. The earthquake cycle. The earthquake prediction. **Seismic Hazard & Risk:** PSHA, DSHA, Site Response, Site amplification and Site Factors. Design Response Spectrum.

Tutorial: Determination of earthquake parameters (epicentre, hypocentre and distances) from seismograms (short period, broadband, long period and strong motion) manually and through HYPO 71. Focal mechanism study through stereographic projection. Peak ground acceleration and modelling. Source parameterisation.

Space for entering the syllabus

Reference Books:

- 1) Introduction to Seismology by Peter M. Shearer
- 2) Quantitative Seismology- Theory and Methods Volume I and II by Keiiti Aki and Paul G. Richards
- 3) Modern Global Seismology by Thorne Lay and Terry C. Wallace
- 4) The Mechanics of Earthquakes and Faulting by Christopher H. Scholz
- 5) Geotechnical Earthquake Engineering by Steven L. Kramer
- 6) The Seismic Design Handbook by Farzad Naeim
- 7) Fault Zone Properties and Earthquake Rupture Dynamics by Eiichi Fukuyama
- 8) Active Faults of the World by Robert Yeats
- 9) Seismic Hazard and Risk Analysis by Robin K. McGuire

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	<u>Classical and Mathematical Seismology:</u> Continuum. Elastic deformation and stress tensors and their relationship, Helmholtz's Theorem. Equation of motion in an elastic medium. Generalized Hooke's Law. Wave surfaces, rays and their properties. Anisotropy and its effects on rays. Rotational, dilatational, irrotational and equivolumnal waves.	06
2.	Velocities and motions associated with body waves. Amplitude and energy transport. Reflection and refraction of elastic waves at solid-solid and liquid-liquid interfaces. Head waves. Inhomogeneous & Evanescent wavefronts and bounded waves.	10
3.	Development of Wave equation solutions in terms of rays & Eikonal equations.	04
4.	Surface Waves: Rayleigh and Love waves.	04
5.	<u>Earthquake Seismology: Introduction:</u> Historical development of seismology. Seismic sources, earthquake sources involving shear faulting. Quantification of earthquakes. Earthquake distributions. Global faulting patterns and rupture models. Internal structure of the earth. The framework and mechanism of plate tectonics, ocean ridges and continental rifts, transform and transcurrent faults, subduction zones, mountain ranges.	03
6.	Seismometry: Inertial pendulum systems. Electromagnetic instruments. Global Networks. Force-	03

	feedback instruments and digital global networks. Seismic arrays and regional networks.	
7.	Seismogram interpretation: Nomenclature. Traveltime curves, locating earthquakes by single and multi station and inverse methods. Determination of earth structure: Seismic earthquake tomography. Joint hypocenter determination (JHD).	03
8.	Seismic sources: Faulting sources, elastostatics, double couple hypothesis, elastodynamics. Seismic moment tensor. Focal mechanisms and fault plane solutions.	03
9.	Earthquake kinematics and dynamics: One-dimensional Haskell source model, source spectrum, stress drop and rupture velocity. Magnitude scales, seismic energy and magnitude. Aftershocks and fault area. Earthquake statistics.	03
10.	Seismotectonics: Divergent, transcurrent and convergent boundaries. Intraplate earthquakes. The earthquake cycle. The earthquake prediction.	03
11.	Seismic Hazard & Risk: PSHA, DSHA, Site Response, Site amplification and Site Factors. Design Response Spectrum.	03
12.	Tutorial: Determination of earthquake parameters (epicentre, hypocentre and distances) from seismograms (short period, broadband, long period and strong motion) manually and through HYPO 71. Focal mechanism study through stereographic projection. Peak ground acceleration and modelling. Source parameterisation.	03
Total number of hours		48